Md. Kaif Faiyaz

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Education

- **Indian Institute of Technology Kanpur** *2019 - 2024 Kanpur, India*
  BS-MS Dual Degree in Physics; Minor in Industrial Management & Engineering
  – CGPA: 8.42/10 (BS), 8.9/10 (MS)

Research Paper

- **Multihead Adaptive Genetic Algorithm in Non-Stationary Environments** *May'24-Nov'25(SUBMITTED)*
  with Prof. Tushar Sandhan ***(under review at IEEE Transactions on Cybernetics)*** **Paper**
  – Recast evolutionary search as a controlled stochastic dynamical system for non-stationary financial environments.
  – Derived warmstart survivorship bounds using Bernstein-Serfling concentration and threshold-exceedance arguments.
  – Analyzed selection dynamics via the Price equation and ECDF-based approximations for intractable fitness probabilities. Validated the framework through ablations on indices and FX, showing robust walk-forward performance improvements.

- **Electrically driven quantum metric Bloch oscillations** *April'23 - April'24(SUBMITTED)*
  with Prof. Amit Kumar Agarwal ***(under review at Physics Review Letter-B)*** **Paper**
  – Developed a modified semiclassical framework to analyze electron wave-packet dynamics under inhomogeneous electric fields, deriving closed-form expressions for real-space oscillations driven by the quantum metric.
  – Conducted analytical and asymptotic derivations of drift and oscillatory transport responses-separating intrinsic vs. extrinsic mechanisms-and established scaling laws for current saturation and differential conductance.
  – Implemented simulations on tilted Dirac models, performing parameter sweeps over field strength, chemical potential, and tilt to validate theoretical predictions and quantify quantum-metric-induced transport behaviors.

Self-directed Research

- **Finite-Sample Theory for Causal Tail Dependence Estimators** *Nov'25 - Present*
  Manuscript in preparation **Project**
  – Derived explicit finite-sample error bounds for nonparametric estimators of the upper tail dependence coefficient, combining Bernstein concentration inequalities with second-order tail bias control.
  – Analyzed rank-based estimators under unknown marginals using DKW-type uniform bounds, quantifying the additional error induced by marginal estimation. Established bias-variance tradeoffs and optimal threshold selection for tail dependence estimation, with validation via copula-based simulations and real hydrological data.

Awards

- **JEE Advanced 2019** Secured **All India Rank 1419 a**mong the 250,000+ screened candidates.
- **JEE Mains 2019** Secured **All India Rank 2286** among the 1.6 million candidates.
- **WBJEE 2019** Secured **All India Rank 63** among the 130,000+ candidates.

Research Projects

- **Modeling of Tail Dependence in Airline Stock Prices** *Aug 2023 - Jan 2024*
  Supervisor: Prof. Arnab Hazra **Project**
  – Modeled joint upper- and lower-tail dependence in Indigo and SpiceJet daily log-returns during COVID-19 using a flexible three-parameter copula with asymmetric Laplace components, combined with GEV-based marginal transformations and local likelihood estimation for time-varying dependence.
  – Performed preprocessing via GEV-based uniformization of return-rate series with spline-selected kernel parameters using bootstrap validation, enabling stable inference of dynamic tail-dependence coefficients and asymptotic (in)dependence classes.
  – Developed a hierarchical Bayesian model for time-varying $(\delta_l, \delta_u, \rho)$ using Kronecker-structured multivariate normal priors; derived Gibbs updates for Θ and β, implemented Metropolis-Hastings samplers for ϕ and Σ, and analyzed numerical stability issues arising from near-singular correlation matrices.

- **Multi-Company-Multi-Period Demand Response Management in Smart Grid** *Sep 2022 - Nov 2022*
  Supervisor: Prof V. Ramani(IME636A)[reproduced original paper] **Project**
  – Formulated a multi-period, multi-company Stackelberg game for demand-response management, deriving closed-form optimal consumer demand, company pricing, and Nash-equilibrium power allocation under constraints.
  – Implemented a distributed price-update algorithm enabling to compute equilibrium strategies.

– Reproduced case studies using EcoGrid EU and Danish Smart Grid datasets, showing reduction in consumer payments by 11-36% , while also validating theoretical predictions on power allocation and equilibrium pricing.

- **Application of Weibull Distribution & Weibull Process** *Sep 2022 - Nov 2022*
  Supervisor: Prof Arnab Hazra(MTH511A) **Project**
  – Implemented a convex mixture of Weibull densities to model multi-seasonal wind-speed distributions, formulating and solving a highly nonlinear constrained MLE problem using Sequential Quadratic Programming (SQP).
  – Performed goodness-of-fit evaluation through empirical CDF construction and two-sample Kolmogorov-Smirnov tests, demonstrating improved fit of the optimized mixture model compared to standard Weibull assumptions.
  – Developed and coded a Monte Carlo EM algorithm for parameter estimation of a Weibull process under left-censored failure-time data, including simulation of missing data via inverse-transform sampling and iterative maximization of the expected log-likelihood.

- **An Efficient Algorithm for Bayesian Nearest Neighbours** *Jan 2022 - April 2022*
  Supervisor: Prof Arnab Hazra(MTH535A)[reproduced original paper] **Project**
  – Developed an efficient Bayesian change-point-based algorithm to compute the posterior distribution of the optimal k for k-Nearest Neighbours, avoiding MCMC and achieving order-of-magnitude faster inference than simulation-based methods.
  – Implemented full probabilistic calibration of neighbourhood size by recursively evaluating change-point probabilities on distance-ordered data, and applied the method to the Ripley dataset for binary classification.
  – Conducted a performance benchmarking study comparing Bayesian k selection with naïve k-NN across 1-125 neighbour regimes, demonstrating a reduction in misclassification rate from 0.104 to as low as 0.036.

Coursework

† Awarded **A/A* grade**

Statistical Inference & Data Analysis†, Data Structures & Algorithms, Game Theory†, Stochastic Processes, Bayesian Analysis†, Probability & Statistics, Quantum Mechanics†, Statistical Mechanics, Linear Algebra, Differential Equations, Complex Analysis, Partial Differential Equations,Statistical Physics,Real Analysis & Multivariable Calculus†

Work Experience

- **Aviso AI** *Jul 2024 - Present*
  Conversational Intelligence, Data Scientist-II Remote
  – Developed a privacy-preserving multimodal deception detection system using self-supervised cross-modal and temporal contrastive learning; framed as weakly supervised inference with multiple-instance learning.
  – Designed a decision-support system for product recommendation and pricing using RAG models; incorporated user feedback via preference-based optimization (DPO), enabling iterative updating of model behavior.
  – Proposed an importance-sampling-based optimization framework for prompt selection using LLM-as-a-judge; employed active sampling strategies to reduce variance and improve sample efficiency in model evaluation.
  – Designed a bimodal speech emotion recognition framework combining embeddings with GRU-based temporal modeling; addressed class imbalance using data augmentation and loss reweighting, improving Macro-F1 by 8.03%.

- **Samsung Research Institute Bangalore** *May 2023 - Jul 2023*
  Developer Intern Bangalore, India
  – Automated **ML pipeline deployments** by orchestrating VMs with Ansible, configuring Kubernetes clusters, and managing Docker environments on remote Linux servers, streamlining scalable experimentation.

- **Global Alpha Researcher (Part-time)** *Remote*
  Trexquant LLP Nov-Dec 2024
  – Selected after surpassing benchmark in Hangman Challenge, achieving 62.4% accuracy in character prediction.
  – Worked on Earning's Call Dataset to create profitable strategies over baseline methods in multiple stock universes.

Technical Skills

- **Programming Languages & DevOps-** C++, Python, R, SQL,Docker, Kubernetes, Ansible, Linux
- **Machine Learning & Deep Learning Frameworks-** PyTorch, TensorFlow, Scikit-learn, PEFT, RLHF,GRUs
- **NLP & LLM Frameworks-** LangChain, LangGraph, HuggingFace Transformers, Redis, AstraDB, Neo4j
- **Tools & Utilities-** Git, LaTeX, Monte Carlo simulation, Time-series analysis ,GARCH models, Copula modeling